Lund University

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Checklist, Topology MATM36/FMAP15, Spring 2026

Below is a list of topics that are covered in the course, which may be examined during the oral exam.

The general principle is easy: all sections from Gamelin's book that were covered should be studied in their entirety, that is: Sections 1-7 of Chapter 1, and Sections 1-12 from Chapter 2. In addition, a few pages on countability from Munkres' book, and the excerpt about Stone-Weierstrass' theorem, from Simmons' book are part of the course.

When learning topology, one encounters many definitions and lesser results. Making an exhaustive list would be too boring. Rather, I will mention a few key results that I think are more substantial. The thought is that if you focus on these, you will learn many of the lesser results in the process.

Some proofs denoted by an asterisk (*) are required only for those who aspire for a VG. (The formulations of the results are included for everybody, but the proofs are only required for VG.)

I want to emphasize that the list is preliminary and may be subject to change.

Set theory

(Munkres, Sec 1.6 and 1.7) The concept of cardinality of a set. Basic facts about finite, countable and uncountable sets. In particular, $\mathbb{Z}_+ \times \mathbb{Z}_+$ is countable, but the power set $\mathcal{P}(\mathbb{Z}_+)$ as well as the infinite product $\prod_1^\infty \{0,1\}$ (consisting of all sequences $(x_j)_1^\infty$ with $x_j \in \{0,1\}$) are uncountable.

1. Metric spaces

(Gamelin, Section 1.1-1.7.) Fundamental notions include neighbourhood, adherent point, convergent sequence, Cauchy sequence, completeness, dense and nowhere dense subsets, equivalent metrics, homeomorphisms, compactness, separable metric space, base of open sets, axiom of second countability, uniformly convergent and uniformly Cauchy sequences of functions from some set S with values in a metric space, continuity of functions between metric spaces. Normed space, Banach space, bounded linear map, operator norm.

I will now list some of the more substantial results from the theory of metric spaces. Note that the list is not exhaustive, but rather intended as a rough guideline. There are many other results that could in principle be brought up.

- Baire's theorem 2.6.
- Theorem 4.1 describing basic open sets for (finite) products of metric spaces.

- Theorem 5.1 giving equivalent conditions that a metric space be compact. This is rather involved and long, but you should be prepared to give an outline of the main steps in the proof and sketch proofs of some of the implications. (For VG you should be prepared to answer somewhat more detailed questions.)
- Formulate and prove Heine-Borel's theorem 5.5, classifying compact subsets of $\mathbb{R}^n$.
- The uniform continuity theorem 6.3.
- Theorem 7.5 in Chap 1: A linear operator between normed spaces is bounded if and only if it is continuous.
- (*) Theorem 7.8: Principle of uniform boundedness.

2. Topological spaces

(Gamelin, Section 2.1-2.12). The basic definitions (neighbourhoods, adherent points, convergent sequences, and so on) go without saying. The separation axioms (T_1)-(T_4), in particular Hausdorff and normal topological spaces. Equicontinuous families of functions. Local compactness and 1-point compactification. Connectedness and path-connectedness, connected component and path-connected component. Product topology. Axiom of choice. Zorn's lemma.

- Theorem 3.4: a uniformly convergent limit of continuous functions is continuous.
- Theorem 4.2: the condition that a family of sets form a basis for some topology.
- Theorem 4.3: Lindelöf's principle.
- Theorem 5.1: metric spaces are normal.
- Theorem 5.3: Urysohn's lemma.
- (*) Theorem 5.4: Tietze's extension theorem.
- Theorem 6.5: Compact Hausdorff spaces are normal.
- Theorem 6.6: A continuous image $f(X)$ of a compact space X is compact.
- Theorem 6.7: A continuous bijection from a compact space to a Hausdorff space is a homeomorphism.
- Arzela-Ascoli's theorem is contained in Exercises 8,9,10 in Section 6 in Gamelin's book. The proof is presented in detail during a lecture.
- Formulate Theorem 7.1 on the 1-point compactification of a locally compact space. (The proof is not required.)
- Theorem 8.1-8.4 on connectedness as well as 9.1-9.3 on path-connectedness.
- Section 10 on finite product spaces: all theorems are relevant.
- Section 11: The axiom of choice and Zorn's lemma (which is an axiom as far as we are concerned).
- Section 12: (*) Tychonoff's theorem.

The Stone-Weierstrass approximation theorems

(Excerpt from the book by Simmons). Subalgebras of $C(X, \mathbb{R})$ and $C(X, \mathbb{C})$ where X is a compact Hausdorff space. Especially subalgebras that separate points of X; in the case of $C(X, \mathbb{C})$ they should also be invariant under complex conjugation.

- Formulate and prove the real Stone-Weierstrass theorem. You may freely use (without proof) the classical Weierstrass approximation theorem, that a continuous function on a compact interval $[a, b]$ can be uniformly approximated by polynomials as well as we please.
- Formulate the complex Stone-Weierstrass theorem and explain briefly how it follows from its real counterpart.